MPG No. 5.1.12 $2/m$ [Type I, monoclinic] [M tensor]

* Rank 0 tensor. * Rank 1 tensor.

$$\begin{bmatrix} 0 & M_y & 0 \end{bmatrix}$$

$$M_y = \mathbb{M}_y^{(1)}$$

* Rank 2 tensor (s). * Rank 2 tensor (a). * Rank 3 tensor (ss).

$$\begin{bmatrix} 0 & M_{xxy} & 0 \\ 0 & M_{yyy} & 0 \\ 0 & M_{zzy} & 0 \\ M_{yzx} & 0 & M_{zzy} \\ 0 & M_{yzx} & 0 \\ M_{xxy} & 0 & M_{yzx} \end{bmatrix}$$

$$M_{xxy} = \mathbb{M}_y^{(1)} - \mathbb{M}_y^{\alpha(1)} - \mathbb{M}_y^{\beta(1)}$$
$$M_{yyy} = 3\mathbb{M}_y^{(1)} + 2\mathbb{M}_y^{\alpha(1)}$$
$$M_{zzy} = \mathbb{M}_y^{(1)} - \mathbb{M}_y^{\alpha(1)} + \mathbb{M}_y^{\beta(1)}$$
$$M_{yzx} = \mathbb{M}_{xyz}^{(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & M_{xxy} & 0 \\ 0 & 0 & 0 \\ 0 & M_{zzy} & 0 \\ M_{yzx} & 0 & -\frac{M_{zzy}}{2} \\ 0 & M_{zxy} & 0 \\ -\frac{M_{xxy}}{2} & 0 & -M_{yzx} - M_{zxy} \end{bmatrix}$$

$$M_{xxy} = 2\mathbb{M}_y^{(2)} - 2\mathbb{T}_{xz}^{(1)}$$
$$M_{zzy} = 2\mathbb{M}_y^{(2)} + 2\mathbb{T}_{xz}^{(1)}$$
$$M_{yzx} = 3\mathbb{T}_u^{(1)} + \mathbb{T}_v^{(1)}$$
$$\frac{M_{zzy}}{2} = \mathbb{M}_y^{(2)} + \mathbb{T}_{xz}^{(1)}$$
$$M_{zxy} = -3\mathbb{T}_u^{(1)} + \mathbb{T}_v^{(1)}$$
$$\frac{M_{xxy}}{2} = \mathbb{M}_y^{(2)} - \mathbb{T}_{xz}^{(1)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} M_{yzx} & 0 & M_{yzz} \\ 0 & M_{zxy} & 0 \\ M_{xyx} & 0 & M_{xyz} \end{bmatrix}$$

$$M_{yzx} = \mathbb{T}_0^{(1)} - \mathbb{T}_u^{(2)} + \mathbb{T}_v^{(2)}$$
$$M_{yzz} = -\mathbb{M}_y^{(3)} + \mathbb{T}_{xz}^{(2)}$$
$$M_{zxy} = \mathbb{T}_0^{(1)} - \mathbb{T}_u^{(2)} - \mathbb{T}_v^{(2)}$$
$$M_{xyx} = \mathbb{M}_y^{(3)} + \mathbb{T}_{xz}^{(2)}$$
$$M_{xyz} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_u^{(2)}$$

* Rank 4 tensor (sss). * Rank 4 tensor (ssa). * Rank 4 tensor (aas). * Rank 4 tensor (aaa). * Rank 4 tensor (sa). * Rank 4 tensor (as). * Rank 4 tensor (s). * Rank 4 tensor (a). * Rank 4 tensor (t).